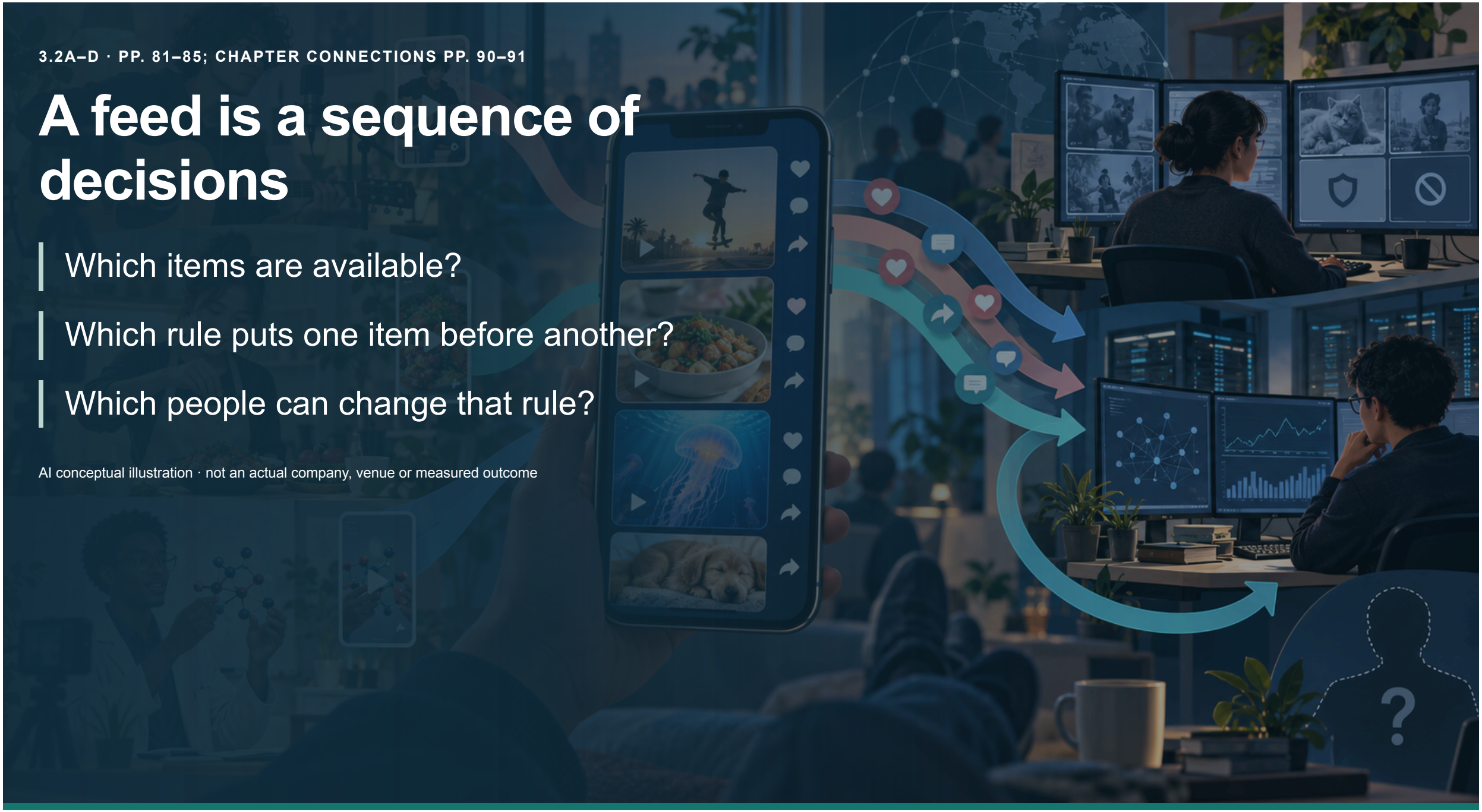
3.2A–D · PP. 81–85; CHAPTER CONNECTIONS PP. 90–91

A feed is a sequence of decisions

- Which items are available?
- Which rule puts one item before another?
- Which people can change that rule?



AI conceptual illustration · not an actual company, venue or measured outcome

An algorithm

> A precise sequence of instructions for a task

> Input: the information available to the rule

> Output: the result produced by the instructions

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 81–82.

Characteristics of a usable algorithm

> Unambiguous: each instruction has one clear meaning

> Finite: the procedure eventually stops

> Well defined and feasible: inputs, outputs and steps are possible

Variables hold values

- A variable is a named place for a value
- Its value may change as the rule runs
- A trace records those changes in order

Step	Instruction	total
Start	Set total to 0	0
Item A	Add 1	1
Item B	Add 1	2

Conditionals choose a branch

- IF a test is true, follow one branch
- ELSE, follow the other branch
- The boundary value matters

```
IF duration <= 60
    include item
ELSE
    exclude item
```

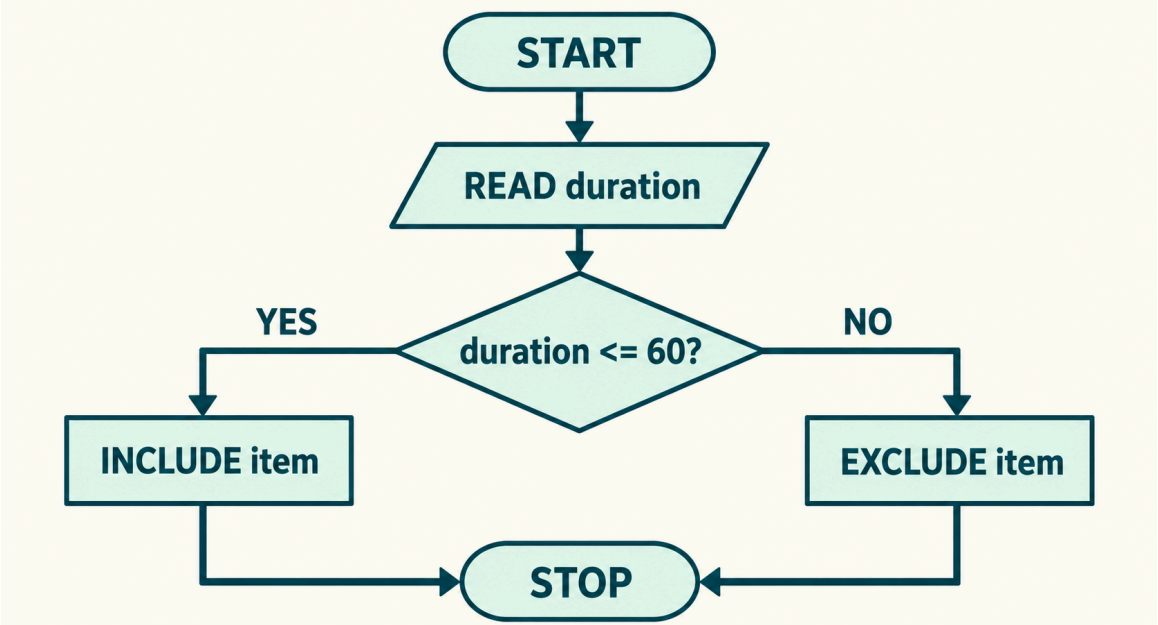
Loops repeat a set of instructions

- FOR each item, apply the same test
- Update the trace after each item
- Stop when the finite list ends

```
total = 0
FOR each item in the list
    IF duration <= 60
        total = total + 1
OUTPUT total
```

One procedure, several representations

- Natural language explains a procedure in words
- A flowchart makes sequence and branches visible
- Code expresses instructions in a programming language



Original teaching diagram · relationships explained in transcript

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 83.

Algorithm, pseudocode and code

> Algorithm: the underlying procedure

> Pseudocode: a readable planning representation

> Code: instructions written for a language and runtime

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 83–84.

Searching finds a target

- Linear search checks records one at a time
- Stop at the first match, or at the end
- A failed search must still terminate

Comparison	Record	Match B3?
1	B4	No
2	B1	No
3	B3	Yes: stop

Sorting changes the order

- Sort arranges existing items by a stated key
- Bubble sort compares adjacent values
- One pass may leave part of the list unsorted

State	Values	Action
Start	3, 1, 2	Compare first pair
After swap	1, 3, 2	Compare next pair
After swap	1, 2, 3	End this pass

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 84.

Filtering and counting

- Filter: keep records that meet a condition
- Count: report how many records meet it
- Sort: order records, including those a filter might remove

Task	Output
Filter loans >= 10	A and C
Count matches	2 records
Sort all, descending loans	C, A, B

Efficiency needs a measure

> Possible measures: comparisons, time or memory

> Input size and assumptions affect the result

> Fast execution and a fair decision are separate tests

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 81, 84–86.

Prioritizing turns a goal into an order

- A ranking score combines chosen signals
- Weights determine how strongly each signal counts
- A tie-break makes equal scores reproducible

Item	Completion c	New topic m
M1	0.8	1
M2	0.9	0
M3	0.5	0

A changed objective: prediction

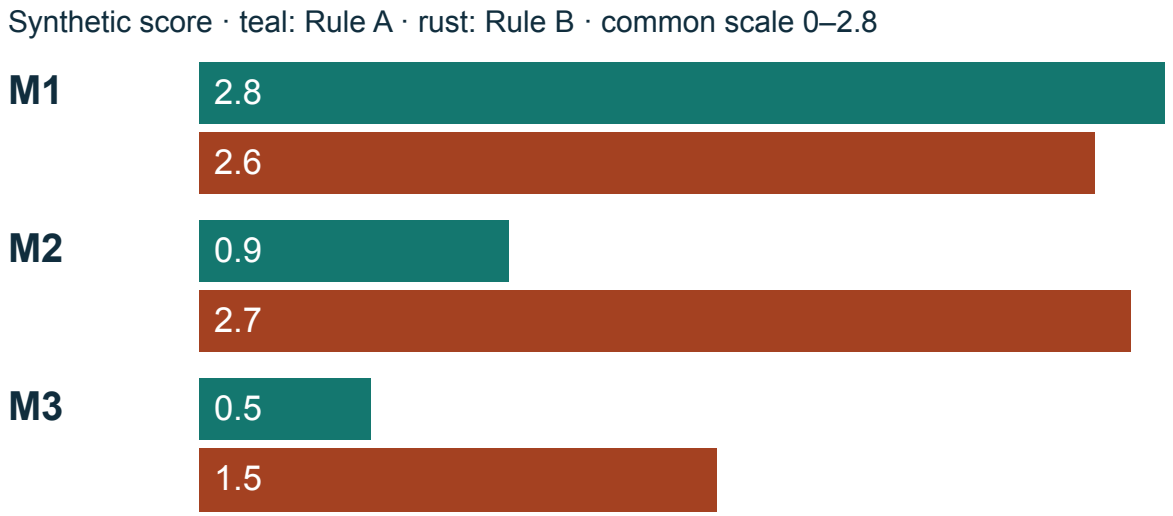
> Rule A: score = c + 2m

> Rule B: score = 3c + 0.2m

> Use the same inputs to compare the rules

A changed objective: worked result

- Rule A ranks M1 first
- Rule B ranks M2 first
- The changed weights alter whose item receives exposure



Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 85.

Classifying and associating

> Classifying assigns records to a category

> Associating identifies patterns of items occurring together

> A pattern can guide a decision without proving its cause

Rules and learned models

> A hand-written rule specifies the calculation

> A learned model uses data to adjust internal parameters

> Both depend on chosen data, objectives and evaluation

IB Digital Society guide, local April 2026 revision, sections 3.1H, 3.2A–E and 4.2A–D. Assessment authority. Textbook connection: pp. 85, 91.

The complete decision pipeline

- Collect and validate the input
- Apply eligibility rules and calculate scores
- Order results, present them and record the response

1. Collect/validate →

2. Filter eligible items →

3. Calculate score →

4. Rank and display

A response after display can become a later input. The rule and objective must be stated, not guessed.

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 84–85, 91.

Independent transfer: a library queue

> A school library prioritizes reservation requests

> Propose one input, one condition and a stopping rule

> Explain a possible loss caused by your chosen priority

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 81–85.

Chapter 3.2: the knowledge to carry forward

- Describe characteristics and components
- Trace a representation and explain its use
- Evaluate the decisions and dilemmas behind a result

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 81–91.